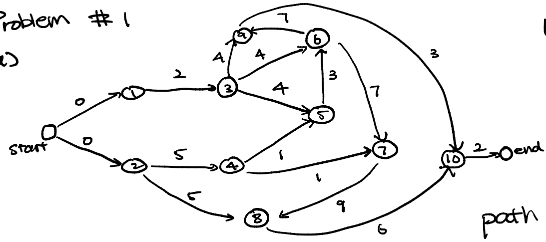
Optimization HW4

Problem #1

(a)



We can build a acyclic network $N(V,A)$ where:

Nodes: all activities with start and end nodes

Arcs: (s,t) of cost 0 if i has no prereq

(i,t) of cost c_i if i is not prereq

(i,j) of cost c_i when i is prereq of j

path from start → end is a chain of mutually dependent events. The cost is the sum of the durations on the chain. The longest path of start → end is the completion time.

(b) modify the graph from part (a) s.t. all the cost values on each arc transforms from $c_i \to -c_i$. The longest path in the network in part (a) becomes shortest path in the negated-cost network. Since the project network is acyclic, there is no negative-cost cycle.

(c) shortest path problem ⟶ Min cost flow

- set capacities for all arcs $u_a = 1$
- keep the cost as negative costs
- there is a flow variable for each arc x_a for $a \in A$

formulation of min flow is:

LP: $\min \sum_{a \in A} x_a c_a$

at the start node: $\sum_{a \in \delta^+(s)} x_a - \sum_{a \in \delta^-(s)} x_a = 1$

at the end node: $\sum_{a \in \delta^+(t)} x_a - \sum_{a \in \delta^-(t)} x_a = -1$

all other nodes: $\sum_{a \in \delta^+(v)} x_a - \sum_{a \in \delta^-(v)} x_a = 0$

constraints:

$0 \le x_a \le 1$ for all $a \in A$

using gurobi to compute,

the project completion time is $T^* = 33$

two longest chains are $1 \to 3 \to 5 \to 6 \to 7 \to 8 \to 10$

$2 \to 4 \to 5 \to 6 \to 7 \to 8 \to 10$

(d) A activity i is critical if it lies on at least one longest start → end path. where its slack time is zero

- Forward time $F(i)$: sort activities in descending order of longest time to finish i
- Backward longest time $B(i)$: sort activities in ascending order of longest time starting at i to end of the project.
- Project duration:

$T^* = \max(F_i)$ This gives the overall project completion time

- Compute Slackness

For each activity i, define $\text{Slack}(i) = T^* - (F(i) + B(i) - c_i)$

Activity i is critical if $\text{Slack}(i) = 0$

Results :

For the problem above, critical activities are 1,2,3,4,5,6,7,8,10

only 9 is non-critical

Problem #2

(a) day 1 day 2 day 3

$$d_1 \xrightarrow[y_1]{q_1} d_2 \xrightarrow[y_2]{q_2} d_3$$

$$c_1 \cdot x_1 \qquad c_2 \cdot x_2 \qquad c_3 \cdot x_3$$

IP:

let x_i = number of units shipped from warehouse to store on day i

$i \in \{1,2,3\}$

let y_j = number of units stored from day j to day $j+1$ $j \in \{1,2\}$

objective function:

$$\min: \sum_{i=1}^{3} x_i \cdot c_i + y_1 \cdot q_1 + y_2 \cdot q_2$$

constraints :

$$x_1 = d_1 + y_1 \qquad x_i \geq 0 \quad \forall i \in \{1,2,3\}$$
$$x_2 = d_2 + y_2 - y_1 \qquad y_j \geq 0 \quad \forall j \in \{1,2\}$$
$$x_3 = d_3 - y_2$$

(b) based on the updated function for q_2. we first add new variables are replace y_2 with y_{2a}, y_{2b}, y_{2c} which represents the 3 parts of y. we also add $u_b, u_c \in \{0,1\}$ as binary values that open and close b and c

the objective function changes to :

$$\min \ x_1 c_1 + x_2 c_2 + x_3 c_3 + y_1 q_1 + 10 y_{2a} + 8 y_{2b} + 5 y_{2c}$$

add constraints :

$$y_2 = y_{2a} + y_{2b} + y_{2c}$$
$$0 \leq y_{2a} \leq 10$$
$$10 u_b \leq y_{2a}$$
$$4 u_c \leq y_{2b}$$
$$0 \leq y_{2b} \leq 4 u_b$$
$$0 \leq y_{2c} \leq C u_c$$
$$u_b, u_c \in \{0,1\}$$

choose C as upper band that is larger than 14

(c) we add variables: $y_1^R = z \geq 0$ day 1 stock at discount rate

$y_1^N = z \geq 0$ day 1 stock at normal rate

$z \in \{0,1\}$ switch to determine if we rent or not

replace the $y_1 q_1$ section with $\frac{q_1}{3} y_1^R + q_1 y_1^N + kz$

constraints :

$$y_1 = y_1^R + y_1^N$$
$$y_1^R \leq Cz$$
$$y_1^N \leq C(1-z)$$
$$z \in \{0,1\}$$

Choose C as valid upper band

there will be scenarios :

- when $z=1$, $y_1^N = 0$ and we pay $k + \frac{q_1}{3} y_1^R$
- when $z=0$, $y_1^R = 0$ and we pay $q_1 \cdot y_1^N$

Problem #3

(a) find IP to minimize number of hires

Hours $H = \{8, 9, 10, 11, 12, 1, 2, 3, 4, 5, 6\}$ denotes slots 8-9, 9-10, ..., 6-7

Clerk Start Times $T_c = \{8, 9, 10, 11, 12, 1, 2, 3, 4\}$

let $x_t^c \in \mathbb{Z}_{\geq 0}$: number of clerk shifts that start at hour $t \in T_c$

Supervisor Start Times $T_s = \{8, 9, 10, 11, 12, 1, 2, 3, 4, 5\}$

let $x_t^s \in \mathbb{Z}_{\geq 0}$: number of supervisor shifts that start at hour $t \in T_s$

objective function:

$$\min \sum_{t \in T_c} x_t^c + \sum_{t \in T_s} x_t^s$$

constraints:

$$\sum_{t \in T_c : h \in \{t, t+1, t+2\}} x_t^c \geq r_h^c$$

$$\sum_{t \in T_s : h \in \{t, t+1\}} x_t^s \geq r_h^s$$

$$x_t^c, x_t^s \in \mathbb{Z} \geq 0$$

hour requirements:

$$r_h^c = (3, 4, 2, 5, 7, 8, 10, 3\cancel{4}, 5, 5)$$
$$r_h^s = (1, 1, 1, 1, 2, 2, 3, 1, 1, 1, 1)$$

(b) minimize the total salary where

Clerk = \$19 Supervisor = \$27

Clerk 3 hour = \$57 Supervisor 3 hour = \$54

objective function becomes: $\min 57 \sum_{t \in T_c} x_t^c + 54 \sum_{t \in T_s} x_t^s$

The optimal solution remains the same. This is because the clerk and supervisor schedule are decoupled, they have to fulfill a coverage constraint. Even with salary, the underlying factor is still to ensure coverage. Therefore the salary is simply a constant value that is multiplied.

(c) Clerks:

- Early if shifts are in $\{8, 9, 10\} \Rightarrow E_t^c = 1$ for $t = \{8, 9, 10\}$; else 0
- Late if third hour is in $\{5, 6\} \Rightarrow L_t^c = 1$ when $t \in \{3, 4\}$; else 0

Supervisor:

- Early if shifts are in $\{8, 9, 10\} \Rightarrow E_t^s = 1$ when $t \in \{8, 9, 10\}$ else 0
- Late if second hour in $\{5, 6\} \Rightarrow L_t^s = 1$ when $t \in \{5, 6\}$ else 0

Costs per start: $C_t^c = 57(1 + 0.1 E_t^c + 0.2 L_t^c)$

$C_t^s = 54(1 + 0.1 E_t^s + 0.2 L_t^s)$

Objective function: $\min \sum_{t \in T_c} C_t^c x_t^c + \sum_{t \in T_s} C_t^s x_t^s$

with same constraints as part (a)